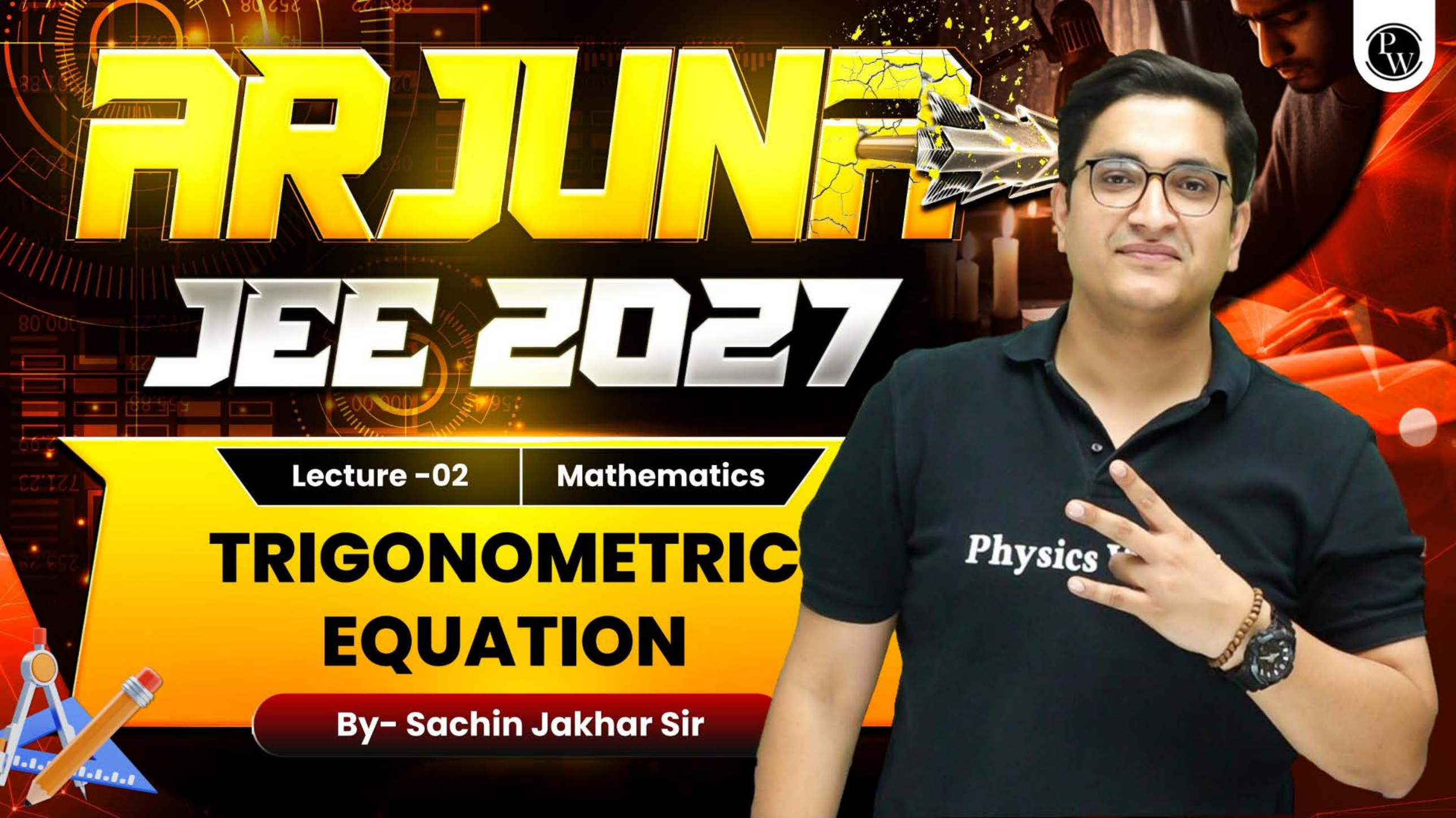
ARJUNA JEE 2027

Lecture -02 | Mathematics

TRIGONOMETRIC EQUATION

By- Sachin Jakhar Sir



Topics *to be covered*

A **Types of Solutions**

B **Four Formula's**

C **Question Practice**

D





LAST CLASS RECAP

Answer the following Questions:

1. Range of $y = a\sin x \pm b\cos x$ is $[-\sqrt{a^2+b^2}, \sqrt{a^2+b^2}]$.
2. Minimum value of: $(a^2\tan^2\theta + b^2\cot^2\theta)$ is $2ab$.
3. Maximum and minimum value of: $(\cos x + \sin x)$ is $[-\sqrt{2}, \sqrt{2}]$.
4. Maximum and minimum value of: $(\cos x + \sin y)$ is $[-2, 2]$.



LAST CLASS RECAP

"Notes"

Answer the following Questions:

5. Range of $y = \cos x + \sin x$, where $x \in \left[0, \frac{\pi}{2}\right]$

$\times$ & $\div$ by $\sqrt{2}$

$$y = \sqrt{2}\left(\frac{1}{\sqrt{2}}\cos x + \frac{1}{\sqrt{2}}\sin x\right)$$

($\frac{1}{\sqrt{2}} = \sin 45^\circ$, $\frac{1}{\sqrt{2}} = \cos 45^\circ$)

$$y = \sqrt{2}\sin\left(x + \frac{\pi}{4}\right)$$

θ

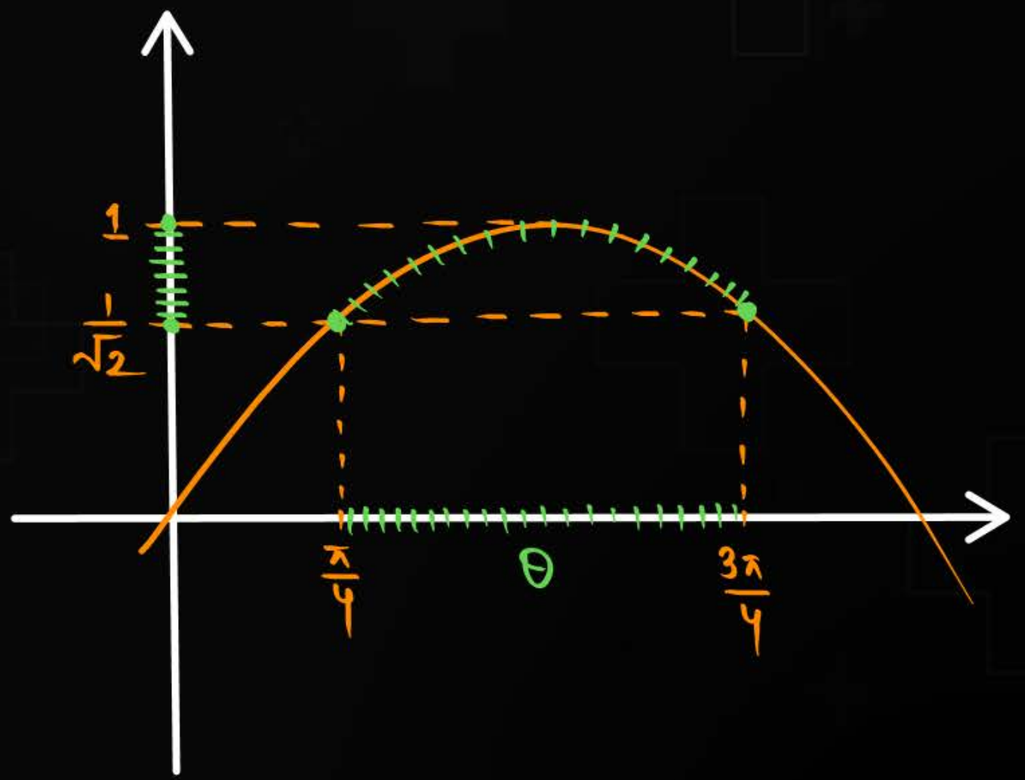
$$x + \frac{\pi}{4} \in \left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$$

$$y = \sqrt{2}\sin\theta\,, \quad \theta \in \left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$$

$$y = \sqrt{2}\sin\theta \cdot , \quad \theta \in \left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$$

$$\sqrt{2} \times \left[\frac{1}{\sqrt{2}}, 1\right]$$

$$y \in [1, \sqrt{2}]$$ fns



H.W. Discussion



RANGE OF TRIGONOMETRIC FUNCTIONS

Type-02 : Range by completing perfect square

Question

(iii) $y = \sin^2 x + \cos^4 x$

[Ans. : $y \in \left[\frac{3}{4}, 1\right]$]

Question

Find the exact value of

(a) $\sqrt{3}\text{cosec}20^\circ - \sec 20^\circ$ ✓

(b) $\text{cosec}10^\circ - \sqrt{3}\sec 10^\circ$

$$\frac{1}{\sin 10^\circ} - \frac{\sqrt{3}}{\cos 10^\circ} = \frac{\cos 10^\circ - \sqrt{3}\sin 10^\circ}{\sin 10^\circ \cdot \cos 10^\circ}$$

$$\frac{2\sin 20^\circ}{\left(\frac{\sin 20^\circ}{2}\right)} = 4$$

$$2 \times \left(\cos 10^\circ - \sqrt{3}\sin 10^\circ\right)$$

$$2\left(\frac{1}{2}\cos 10^\circ - \frac{\sqrt{3}}{2}\sin 10^\circ\right)$$

$\sin 30^\circ$ $\quad$ $\cos 30^\circ$

$$2\sin(30^\circ - 10^\circ)$$

$$2\sin 20^\circ$$

Question

Find the maximum & minimum value of :

(iii) $y = \log_2\left(\frac{3\sin x - 4\cos x + 15}{10}\right)$

$[-5,5] + 15 \Rightarrow [10, 20]$

$\frac{1}{10} \times [10, 20] \rightarrow [1, 2]$

(iv) $y = 27^{\cos 3x} \cdot 81^{\sin 3x}$

$y = 3^{3\cos 3x} \cdot 3^{4\sin 3x}$

$y = 3^{3\cos 3x + 4\sin 3x}$, $3\cos 3x + 4\sin 3x \in [-5,5]$

$y_{max.} = 3^5 = 243$

$y_{min.} = 3^{-5} = \frac{1}{243}$

$y \in \left[\frac{1}{243}, 243\right]$

$\log_2([1,2]) = [0,1]$ Ans.

Question [IIT-JEE-2010]

Find the maximum & minimum value of :

$$y = \frac{1}{\sin^2\theta + 3\sin\theta\cos\theta + 5\cos^2\theta}$$

$y = \frac{1}{D^r}$

$$D^r = \frac{1}{2} - \frac{\cos 2\theta}{2} + \frac{3}{2}\sin 2\theta + 5\left(\frac{1}{2} + \frac{\cos 2\theta}{2}\right)$$

$$= 3 + \left(\frac{5}{2} - \frac{1}{2}\right)\cos 2\theta + \frac{3}{2}\sin 2\theta$$

$$= 3 + \left(2\cos 2\theta + \frac{3}{2}\sin 2\theta\right) \in \left[-\frac{5}{2}, \frac{5}{2}\right] \longrightarrow D^r \in \left[\frac{1}{2}, \frac{11}{2}\right]$$

$\sqrt{a^2+b^2} = \sqrt{4+\frac{9}{4}} = 5/2$

$\frac{1}{D^r} \in \left[\frac{2}{11}, 2\right] \equiv y$

Acidic Question

If $y=\frac{17+5\sin x+12\cos x}{17-5\sin x-12\cos x}$, then the minimum and maximum values of y are respectively



A $\frac{1}{15}, 15$

B $\frac{2}{15}, \frac{15}{2}$

C $\frac{1}{8}, 8$

D $\frac{4}{17}, \frac{17}{4}$

Critical Thinking Question

$y = 18\sec^2 x + 8\cos^2 x$, find minimum value of y & also find corresponding value of x.



Hint: Type-04 (Reciprocal Relationship)



TRIGONOMETRIC EQUATION



THINK ANGLE CORRESPONDING TO VALUE

S-I: Think acute angle (θ) corresponding to given value.

S-II: According to sign, decide quadrant.

S-III: Rotate θ in required quadrants.

ex: $\tan x = -\sqrt{3}$

II or IV

$\theta = \frac{\pi}{3}$ or 60°

$\pi - \theta$	θ
$3\pi - \theta$	$2\pi + \theta$
$\pi + \theta$	$2\pi - \theta$
$3\pi + \theta$	$4\pi - \theta$

For +ve angles.

$-\pi - \theta$	$-2\pi + \theta$
$-3\pi - \theta$	$-4\pi + \theta$
$-\pi + \theta$	$-\theta$
$-3\pi + \theta$	$-2\pi - \theta$
	$-4\pi - \theta$

For –ve angles.

II: $-\pi - \frac{\pi}{3}$, $\pi - \frac{\pi}{3}$; $-3\pi - \frac{\pi}{3}$, $3\pi - \frac{\pi}{3}$; $-5\pi - \frac{\pi}{3}$, $5\pi - \frac{\pi}{3}$; ...

IV: $2\pi - \frac{\pi}{3}$, $-\frac{\pi}{3}$; $4\pi - \frac{\pi}{3}$, $-2\pi - \frac{\pi}{3}$; $6\pi - \frac{\pi}{3}$, $-4\pi - \frac{\pi}{3}$; ...

THINK ANGLE CORRESPONDING TO VALUE

S-I: Think acute angle (θ) corresponding to given value.

S-II: According to sign, decide quadrant.

S-III: Rotate θ in required quadrants.

$\pi - \theta$ $3\pi - \theta$	θ $2\pi + \theta$
$\pi + \theta$ $3\pi + \theta$	$2\pi - \theta$ $4\pi - \theta$

For +ve angles.

$-\pi - \theta$ $-3\pi - \theta$	$-2\pi + \theta$ $-4\pi + \theta$
$-\pi + \theta$ $-3\pi + \theta$	$-\theta$ $-2\pi - \theta$ $-4\pi - \theta$

For –ve angles.

ex: $\sec x = -2$ → II or III

$\theta = \frac{\pi}{3}$

II: $\pi - \frac{\pi}{3}$, $3\pi - \frac{\pi}{3}$, $-\pi - \frac{\pi}{3}$, $-3\pi - \frac{\pi}{3}$, ...

III: $\pi + \frac{\pi}{3}$, $3\pi + \frac{\pi}{3}$, $-\pi + \frac{\pi}{3}$, $-3\pi + \frac{\pi}{3}$, ...

THINK ANGLE CORRESPONDING TO VALUE

S-I: Think acute angle (θ) corresponding to given value.

S-II: According to sign, decide quadrant.

S-III: Rotate θ in required quadrants.

$\pi - \theta$ $3\pi - \theta$	θ $2\pi + \theta$
$\pi + \theta$ $3\pi + \theta$	$2\pi - \theta$ $4\pi - \theta$

For +ve angles.

$-\pi - \theta$ $-3\pi - \theta$	$-2\pi + \theta$ $-4\pi + \theta$
$-\pi + \theta$ $-3\pi + \theta$	$-\theta$ $-2\pi - \theta$ $-4\pi - \theta$

For –ve angles.

ex: $\cot x = +1$ — I or III

$\theta = \frac{\pi}{4}$

I: $\frac{\pi}{4}$, $2\pi + \frac{\pi}{4}$, $4\pi + \frac{\pi}{4}$, $-2\pi + \frac{\pi}{4}$, $-4\pi + \frac{\pi}{4}$

III: $\pi + \frac{\pi}{4}$, $3\pi + \frac{\pi}{4}$, $-\pi + \frac{\pi}{4}$, $-3\pi + \frac{\pi}{4}$, ⋮



THINK ANGLE CORRESPONDING TO VALUE

S-I: Think acute angle (θ) corresponding to given value.

S-II: According to sign, decide quadrant.

S-III: Rotate θ in required quadrants.

$\pi - \theta$ $3\pi - \theta$	θ $2\pi + \theta$
$\pi + \theta$ $3\pi + \theta$	$2\pi - \theta$ $4\pi - \theta$

For +ve angles.

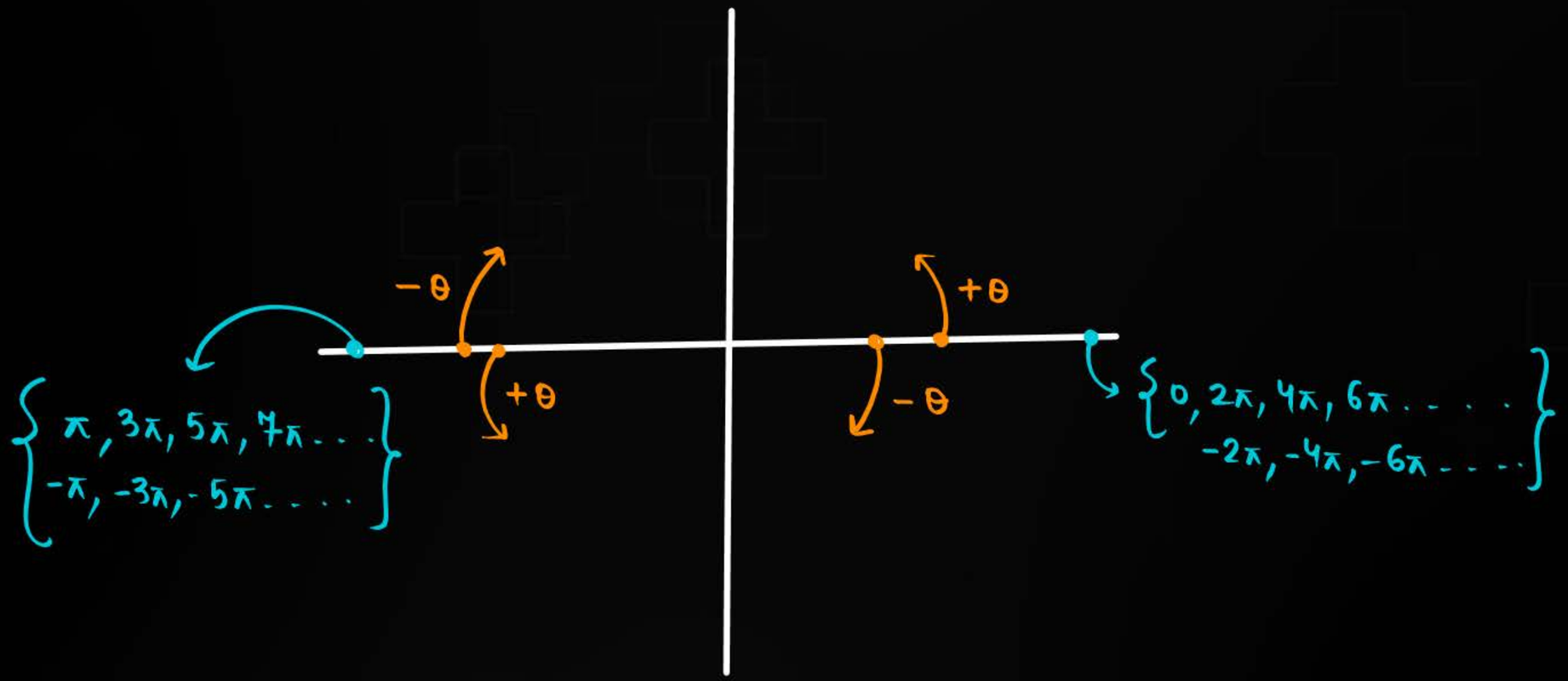
$-\pi - \theta$ $-3\pi - \theta$	$-2\pi + \theta$ $-4\pi + \theta$
$-\pi + \theta$ $-3\pi + \theta$	$-\theta$ $-2\pi - \theta$ $-4\pi - \theta$

For –ve angles.

DIBY

1. $\sin x = -\frac{\sqrt{3}}{2}$
2. $\cos x = -\frac{1}{\sqrt{2}}$
3. $\operatorname{cosec} x = +\sqrt{2}$
4. $\tan x = -\frac{1}{\sqrt{3}}$

Conclusion





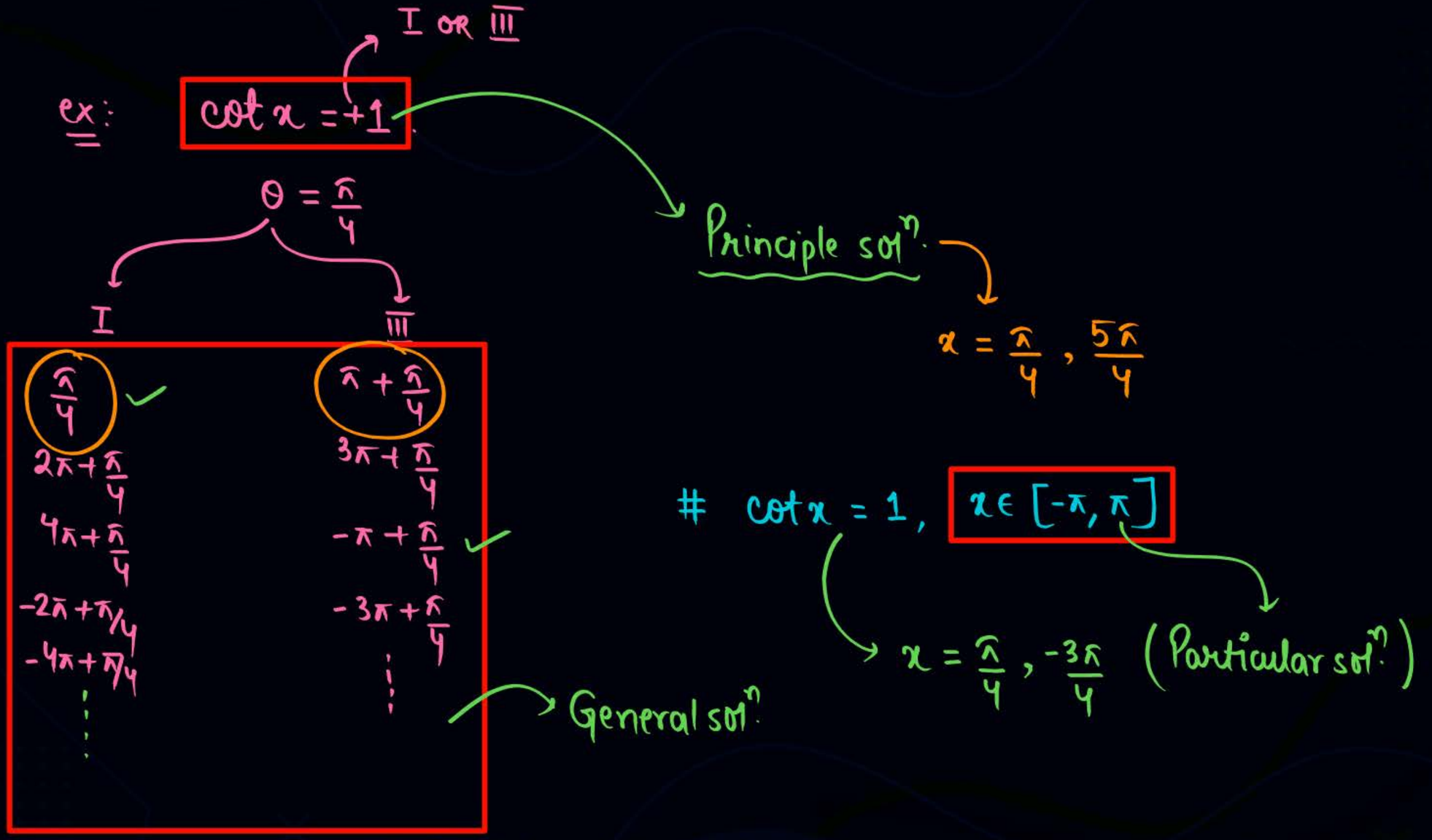
GENERAL SOLUTIONS OF T.E.

Types of Solutions :

General Solution : All angles

Principle Solution: $\theta \in [0, 2\pi)$

Particular Solution : All solutions in the given interval


I or III
ex:
$\cot x = +1$
$\theta = \frac{\pi}{4}$
I
III
$\frac{\pi}{4}$
$2\pi + \frac{\pi}{4}$
$4\pi + \frac{\pi}{4}$
$-2\pi + \pi/4$
$-4\pi + \pi/4$
$\pi + \frac{\pi}{4}$
$3\pi + \frac{\pi}{4}$
$-\pi + \frac{\pi}{4}$
$-3\pi + \frac{\pi}{4}$
Principle soln.
$x = \frac{\pi}{4}, \frac{5\pi}{4}$
General soln.
$\cot x = 1$, $x \in [-\pi, \pi]$
$x = \frac{\pi}{4}, -\frac{3\pi}{4}$ (Particular soln.)

IMPORTANT NOTE

$\sin x = 0 \Rightarrow x = n\pi, n \in I$.

$\cos x = 0 \Rightarrow x =$ odd multiples of $\frac{\pi}{2}$ OR $x = (2n+1)\frac{\pi}{2}, n \in I$.

$\sin x = 1 \Rightarrow x = (4n+1)\frac{\pi}{2}$ OR $(4n-3)\frac{\pi}{2}, n \in I$.

$\sin x = -1$

$\cos x = 1$

$\cos x = -1$

$x = (4n+3)\frac{\pi}{2}$ OR $(4n-1)\frac{\pi}{2}, n \in I$.

$x = 2n\pi, n \in I$

$x = (2n+1)\pi, n \in I$.

IMPORTANT NOTE (RAR)

S. No.	Trigonometric Equation	Principal Solution	General Equation
1.	$\sin\theta = 0$	$\theta = 0, \pi$	$\theta = n\pi, n \in I$
2.	$\sin\theta = 1$	$\theta = \frac{\pi}{2}$	$\theta = (4n+1)\frac{\pi}{2}, n \in I$
3.	$\sin\theta = -1$	$\theta = \frac{3\pi}{2}$	$\theta = (4n-1)\frac{\pi}{2}, n \in I$
4.	$\cos\theta = 0$	$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$	$\theta = (2n+1)\frac{\pi}{2}, n \in I$
5.	$\cos\theta = 1$	$\theta = 0$	$\theta = 2n\pi, n \in I$
6.	$\cos\theta = -1$	$\theta = \pi$	$\theta = (2n+1)\pi, n \in I$
7.	$\tan\theta = 0$	$\theta = 0, \pi$	$\theta = n\pi, n \in I$

Question

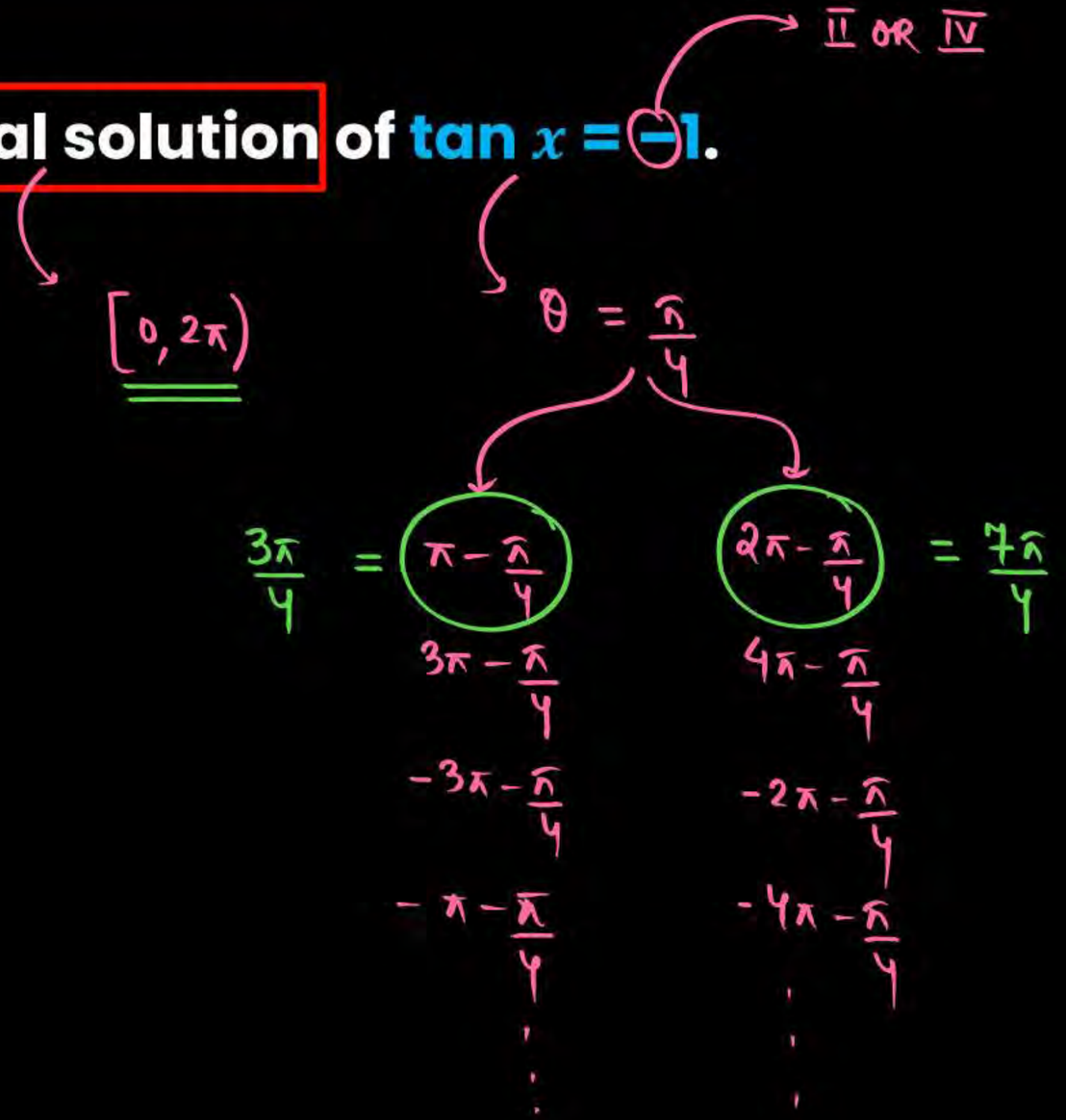
Find the principal solution of $\tan x = -1$.

A $\frac{\pi}{4}, \frac{3\pi}{4}$

B $\frac{3\pi}{4}, \frac{5\pi}{4}$

C $\frac{5\pi}{4}, \frac{7\pi}{4}$

D $\frac{3\pi}{4}, \frac{7\pi}{4}$



Question

Principle solution of $\sin(2\theta) = -\frac{\sqrt{3}}{2}$ is

A $\frac{4\pi}{3}, \frac{5\pi}{3}$

B $\frac{-\pi}{3}, \frac{-2\pi}{3}$

C $\frac{2\pi}{3}, \frac{5\pi}{6}$

D $\frac{2\pi}{3}, \frac{5\pi}{6}, \frac{5\pi}{3}, \frac{11\pi}{6}$

α

$\theta \in [0, 2\pi)$

$2\theta \in [0, 4\pi)$

III OR IV.

$\sin\alpha = -\frac{\sqrt{3}}{2}$, $\alpha \in [0, 4\pi)$

$\frac{\pi}{3}$

III: $\pi + \frac{\pi}{3}$, $3\pi + \frac{\pi}{3}$, ...

IV: $2\pi - \frac{\pi}{3}$, $4\pi - \frac{\pi}{3}$, ...

$2\theta = \alpha = \frac{4\pi}{3}, \frac{10\pi}{3}, \frac{5\pi}{3}, \frac{11\pi}{3}$

$\theta = \frac{2\pi}{3}, \frac{5\pi}{3}, \frac{5\pi}{6}, \frac{11\pi}{6}$

RAPID FIRE ROUND

Find number of solution:

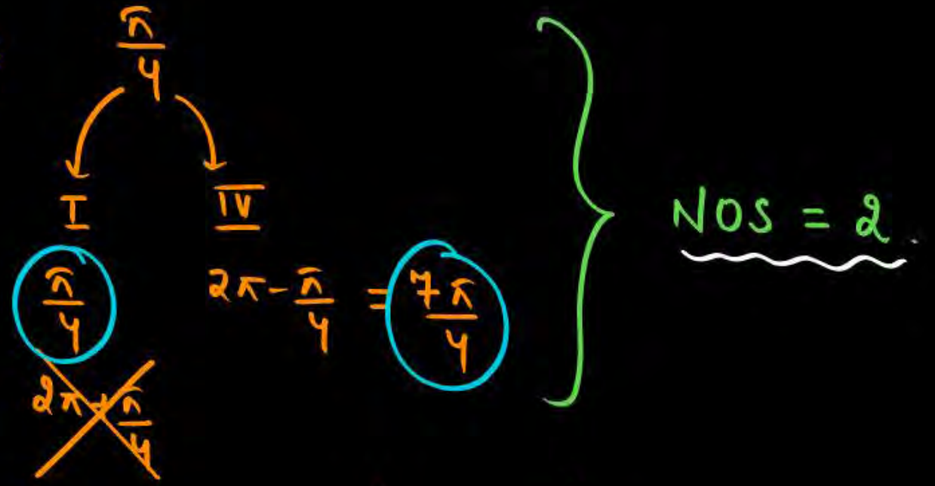
(i) If $x \in [0, 2\pi]$ & $\cos x = \pm\frac{1}{\sqrt{2}}$

(ii) If $x \in [0, 5\pi]$ & $\cos x = \frac{1}{\sqrt{2}}$

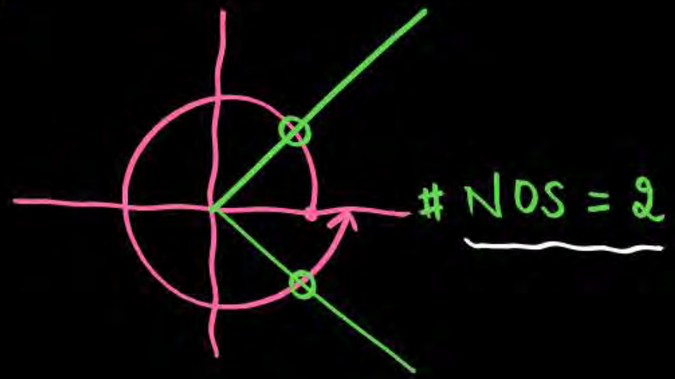
(iii) If $x \in [-\pi, \pi]$ & $\cos x = \frac{1}{\sqrt{2}}$

(iv) If $x \in (-3\pi, 3\pi)$ & $\cos x = -\frac{\sqrt{3}}{2}$

Sumit:



Santosh:



RAPID FIRE ROUND

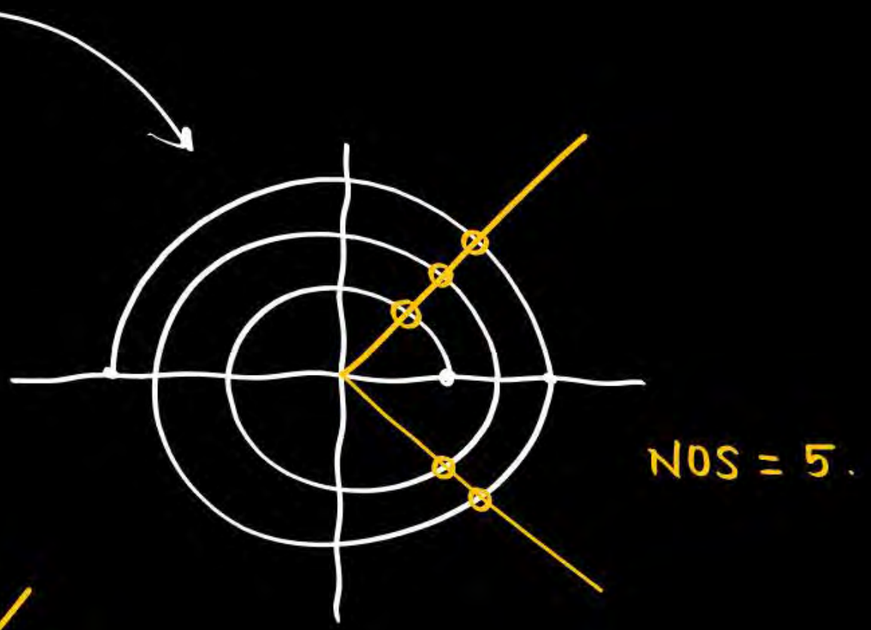
Find number of solution:

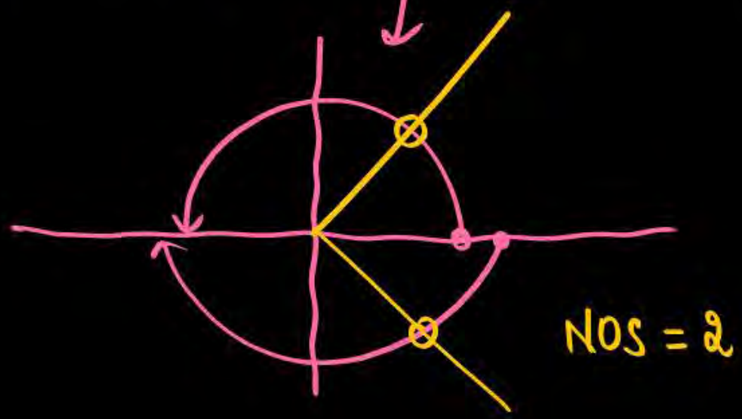
(i) If $x \in [0, 2\pi]$ & $\cos x = \pm\frac{1}{\sqrt{2}}$

(ii) If $x \in [0, 5\pi]$ & $\cos x = \frac{1}{\sqrt{2}}$

(iii) If $x \in [-\pi, \pi]$ & $\cos x = \frac{1}{\sqrt{2}}$

(iv) If $x \in (-3\pi, 3\pi)$ & $\cos x = -\frac{\sqrt{3}}{2}$





RAPID FIRE ROUND

Find number of solution:

(i) If $x \in [0, 2\pi]$ & $\cos x = \pm\frac{1}{\sqrt{2}}$

(ii) If $x \in [0, 5\pi]$ & $\cos x = \frac{1}{\sqrt{2}}$

(iii) If $x \in [-\pi, \pi]$ & $\cos x = \frac{1}{\sqrt{2}}$

(iv) If $x \in (-3\pi, 3\pi)$ & $\cos x = \ominus\frac{\sqrt{3}}{2}$

$(-3\pi, 0] \cup [0, 3\pi)$



RAPID FIRE ROUND

Find number of solution:

(v) If $x \in \left(0, \frac{\pi}{4}\right)$ & $\sin \overset{\alpha}{8x} = \frac{1}{2}$

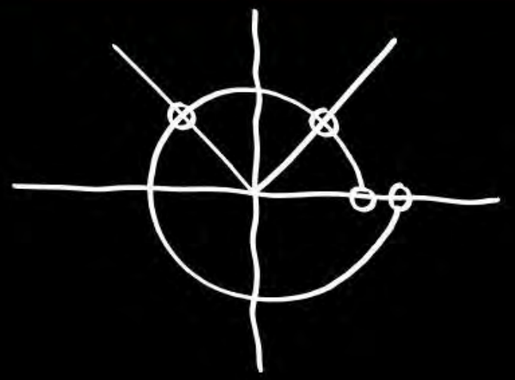
(vi) If $x \in (0, 12)$ & $\tan x = -1$

(vii) If $x \in [0, 6\pi]$ & $\tan 5x = -\frac{1}{3}$

$x \in \left(0, \frac{\pi}{4}\right)$

$8x \in (0, 2\pi)$

$\alpha \in (0, 2\pi)$ & $\sin\alpha = \pm\frac{1}{2}$



NOS = 8

RAPID FIRE ROUND

Find number of solution:

(v) If $x \in \left(0, \frac{\pi}{4}\right)$ & $\sin 8x = \frac{1}{2}$

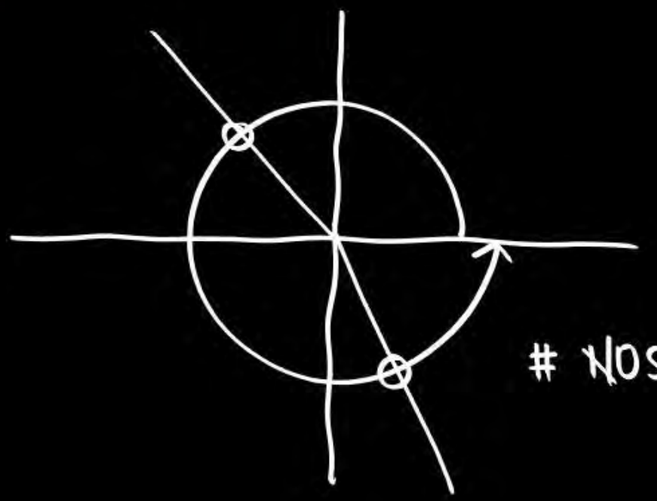
(vi) If $x \in (0, 12)$ & $\tan x = -1$

(vii) If $x \in [0, 6\pi]$ & $\tan \overset{\alpha}{(5x)} = -\frac{1}{3}$

$5x \in [0, 30\pi]$

$\alpha \in [0, 30\pi]$ & $\tan\alpha = \ominus\frac{1}{3}$

II IV



NOS = 15 × 2

= 30.

RAPID FIRE ROUND

Find number of solution:

(v) If $x \in \left(0, \frac{\pi}{4}\right)$ & $\sin 8x = \frac{1}{2}$

(vi) If $x \in (0, 12)$ & $\tan x = -1$

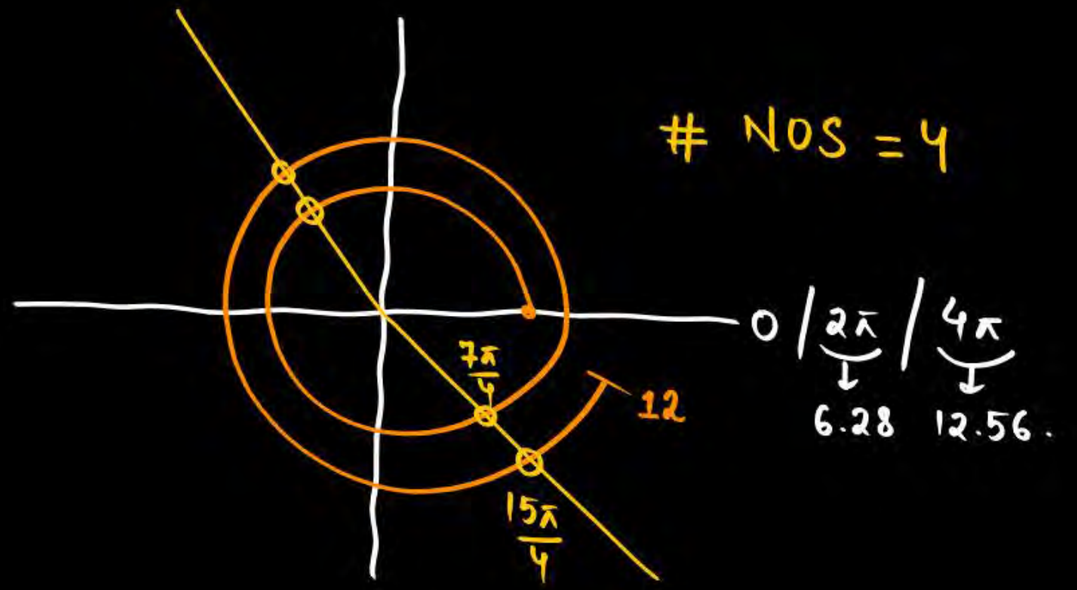
(vii) If $x \in [0, 6\pi]$ & $\tan 5x = -\frac{1}{3}$

$\tan x = -1$

$\frac{\pi}{4}$

$2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$

$4\pi - \frac{\pi}{4} = \frac{15\pi}{4}$



$\frac{15\pi}{4} \leq 12$

$15\pi \leq 48$

Question [JEE Mains-2021]

The number of solutions of the equation $|\cot x| = \cot x + \frac{1}{\sin x}$ in the interval $[0, 2\pi]$ is





FOUR FORMULA'S

(1) $\sin\theta = \sin\alpha \Rightarrow \theta = n\pi + (-1)^n\alpha, (n \in I)$

(2) $\cos\theta = \cos\alpha \Rightarrow \theta = 2n\pi \pm \alpha, (n \in I)$

(3) $\tan\theta = \tan\alpha \Rightarrow \theta = n\pi + \alpha, (n \in I)$

(4) $\left.\begin{array}{l}\sin^2\theta = \sin^2\alpha \\ \cos^2\theta = \cos^2\alpha \\ \tan^2\theta = \tan^2\alpha\end{array}\right\} \Rightarrow \theta = n\pi \pm \alpha, (n \in I)$

$\cos 2\theta = \cos 2\alpha$

$\sin x = \frac{1}{2}$

$\sin(x) = \sin\left(\frac{\pi}{6}\right)$ (θ, α)

$\sin\theta = \sin\alpha$

$\theta = n\pi + (-1)^n \alpha$

$x = n\pi + (-1)^n \frac{\pi}{6}, \quad n \in I$

$n = 0 \Rightarrow x = \frac{\pi}{6}$

$n = 1 \Rightarrow x = \pi - \frac{\pi}{6}$

$n = -2 \Rightarrow x = -2\pi + \frac{\pi}{6}$ - - - - -

$\frac{\pi}{6}$

I	II
$\frac{\pi}{6}$ ✓	$\pi - \frac{\pi}{6}$ ✓
$2\pi + \frac{\pi}{6}$	$3\pi - \frac{\pi}{6}$
$4\pi + \frac{\pi}{6}$	$5\pi - \frac{\pi}{6}$
$-2\pi + \frac{\pi}{6}$ ✓	$-\pi - \frac{\pi}{6}$
$-4\pi + \frac{\pi}{6}$	$-3\pi - \frac{\pi}{6}$
⋮	⋮

Proof :-

$$\sin\theta = \sin\alpha$$

C D

$$\# \quad \sin\theta - \sin\alpha = 0$$

$$2\cos\left(\frac{\theta+\alpha}{2}\right)\sin\left(\frac{\theta-\alpha}{2}\right) = 0$$

$\cos\left(\frac{\theta+\alpha}{2}\right) = 0$ OR $\sin\left(\frac{\theta-\alpha}{2}\right) = 0$

$\frac{\theta+\alpha}{2} = (2n+1)\frac{\pi}{2}$

$\theta = (2n+1)\pi - \alpha$

$\frac{\theta-\alpha}{2} = n\pi$

$\theta = 2n\pi + \alpha$

$$\theta = n\pi + (-1)^n \alpha, \quad n \in I.$$

Proof :-

$$\cos\theta = \cos\alpha$$

$$\cos\theta - \cos\alpha = 0$$

$$-2\sin\left(\frac{\theta+\alpha}{2}\right)\sin\left(\frac{\theta-\alpha}{2}\right) = 0$$

$$\sin\left(\frac{\theta+\alpha}{2}\right) = 0 \quad \text{OR} \quad \sin\left(\frac{\theta-\alpha}{2}\right) = 0$$

$$\frac{\theta+\alpha}{2} = n\pi \quad \cup \quad \frac{\theta-\alpha}{2} = n\pi$$

$$\theta = 2n\pi - \alpha \quad \cup \quad \theta = 2n\pi + \alpha$$

Question

$\tan\frac{3\pi}{4} = \tan\left(\frac{-\pi}{4}\right) = -1$

Write general solution of following equation :

$\tan 3\theta = -1$

$\tan(3\theta) = \tan\left(\frac{-\pi}{4}\right)$ (θ, α)

$\Downarrow$

$3\theta = n\pi + \left(\frac{-\pi}{4}\right)$

$3\theta = \frac{4n\pi - \pi}{4}$

$\theta = (4n-1)\frac{\pi}{12}, \quad n \in I.$

$\tan(3\theta) = \tan\left(\frac{3\pi}{4}\right)$ (θ, α)

$\Downarrow$

$3\theta = n\pi + \frac{3\pi}{4}$

$3\theta = \frac{4n\pi + 3\pi}{4}$

$\theta = (4n+3)\frac{\pi}{12}, \quad n \in I$

same.

$\tan\theta = \tan\alpha$

$\Downarrow$

$\theta = n\pi + \alpha$

Question

The general solution of the following equation : $\cos 3\theta = -\frac{1}{2}$



A $2n\pi \pm \frac{2\pi}{3}$

B $2n\pi \pm \frac{\pi}{3}$

C $\frac{2n\pi}{3} \pm \frac{2\pi}{9}$

D $\frac{2n\pi}{3} \pm \frac{4\pi}{3}$

Question

The general solution of $\tan\left(\frac{2}{3}\theta\right) = \sqrt{3}$.



A $\frac{3n\pi}{2} + \frac{\pi}{2}; n \in I$

B $\frac{n\pi}{2} + \frac{\pi}{2}; n \in I$

C $n\pi \pm \frac{\pi}{2}; n \in I$

D None of these

Question [IIT-JEE-2005]

Let $\theta \in \left(0, \frac{\pi}{4}\right)$ and $t_1 = (\tan\theta)^{\tan\theta}$, $t_2 = (\tan\theta)^{\cot\theta}$, $t_3 = (\cot\theta)^{\tan\theta}$, $t_4 = (\cot\theta)^{\cot\theta}$, then

Sol.

A $t_1 > t_2 > t_3 > t_4$

B $t_4 > t_3 > t_1 > t_2$

C $t_3 > t_1 > t_2 > t_4$

D $t_2 > t_3 > t_1 > t_4$

Question [JEE Mains-2021]

The range of the function

$f(x) = \log_{\sqrt{5}}\left(3 + \cos\left(\frac{3\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} - x\right) - \cos\left(\frac{3\pi}{4} - x\right)\right)$ is

A $\left[\frac{1}{\sqrt{5}}, \sqrt{5}\right]$

B $[0, 2]$

C $(0, \sqrt{5})$

D $[-2, 2]$

It is never about end result

It's always about Journey

Thank You

#futureIITians